

04-28-2025

$$k^{sp} = \left(\frac{\alpha}{n} \right)^{\frac{1}{1-\alpha}}$$

$$k^{ce} = \left[\frac{1-\alpha-\alpha\beta(1+n)}{(1+\beta)(1+n)} \right]^{\frac{1}{1-\alpha}}$$

Part 3:

$$\frac{B_{t+1}}{L_t} = b; \quad B_{t+1} + \pi_t = B_t(1+r_t)$$

$$\frac{B_{t+1}}{L_t} + \frac{\pi_t}{L_t} = \frac{B_t(1+r_t)}{L_t} \cdot \frac{L_{t+1}}{L_{t+1}}$$

$$b + \gamma_t = \frac{b}{1+n} (1+r_t) \Rightarrow b \frac{r_t - n}{1+n} = \gamma_t$$

$$\min \{c^y, \beta c^o\} \rightarrow \max_{c^y, c^o, s}$$

$$w - s - \gamma = \beta s(1+r)$$

$$\hookrightarrow c^y = \beta c^o$$

$$w - \gamma = s(1+\beta(1+r)) \Rightarrow s = \frac{w - \gamma}{1+\beta(1+r)}$$

MCC:

$$K_{t+1} = s_t L_t \Rightarrow k(1+n) = s \quad (\text{in case w/o gov.})$$

$$\text{with gov't: } k(1+n) + b = s$$

$$k(1+n) + b = \frac{w - \gamma}{1+\beta(1+r)} = \frac{w}{1+\beta(1+r)} - \frac{1}{1+\beta(1+r)} b \stackrel{=0}{\frac{r-n}{1+n}}$$

$$\text{if } k = k^{sp} \Rightarrow r = n$$

$$k(1+n) + b = \frac{(1-\alpha)k^\alpha}{1+\beta(1+\alpha k^{\alpha-1})} \Rightarrow b = \frac{(1-\alpha)k^\alpha}{1+\beta(1+\alpha k^{\alpha-1})} - k^{sp}(1+n)$$

Substitute for all factor prices

Social Security Case

$$c^y = w - s - \tau$$

$$c^o = s(1+r) + p$$

s.t.

$$p = \tau(1+n)$$

$$w - s - \tau = \beta(s(1+r) + p)$$

$$w - \tau - \beta p = s(1 + \beta(1+r))$$

$$s = \frac{w - \tau - \beta p}{1 + \beta(1+r)} = \frac{w - \tau - \beta \tau(1+n)}{1 + \beta(1+r)}$$

MCC

$$k(1+n) = s$$

$$k(1+n) = \frac{w - \tau(1 + \beta(1+n))}{1 + \beta(1+r)} = \frac{w}{1 + \beta(1+r)} - \tau \frac{(1 + \beta(1+n))}{1 + \beta(1+r)}$$

$$k(1+n) = \frac{(1-\alpha)k^\alpha}{1 + \beta(1+\alpha k^{\alpha-1})} - \tau \Rightarrow \tau = \frac{(1-\alpha)k^\alpha}{1 + \beta(1+\alpha k^{\alpha-1})} - k(1+n)^{sp}$$

$\hookrightarrow r=n$ under k^{sp}

Coding:

- add premium to MCC

- w/ survival uncert:

surv. premium

$$\text{bequest} = (1 - \text{surv})s / 1+n$$